

Exercise 1E-2

The dihedral angle between two planes is the same as the angle between their normal vectors.

Normal vectors:

$$\mathbf{n}_1 = (2, -1, 1), \quad \mathbf{n}_2 = (1, 1, 2)$$

$$\begin{aligned}\cos \theta &= \frac{\mathbf{n}_1 \cdot \mathbf{n}_2}{\|\mathbf{n}_1\| \|\mathbf{n}_2\|} \\ &= \frac{2(1) + (-1)(1) + 1(2)}{\sqrt{2^2 + (-1)^2 + 1^2} \sqrt{1^2 + 1^2 + 2^2}} \\ &= \frac{3}{\sqrt{6}\sqrt{6}} = \frac{1}{2} \\ \theta &= 60^\circ\end{aligned}$$

Exercise 1E-3

a)

$$x = 1 + 2t, \quad y = -t, \quad z = -1 + 3t$$

b)

$$x = 2 + t, \quad y = -1 - t, \quad z = -1 + 2t$$

c)

The direction vector of the line must be parallel to the plane, therefore it must be perpendicular to the normal vector.

$$x = 1 + at, \quad y = 1 + bt, \quad z = 1 + ct$$

where

$$a + 2b - c = 0$$

and a, b, c are not all zero.

Exercise 1E-4

The line has direction vector

$$\vec{PQ} = (2, -1, 1)$$

Parametric equations of the line:

$$x = t, \quad y = 1 - t, \quad z = 2 + t$$

Substitute into the equation of the plane:

$$2t + 4(1 - t) + (2 + t) = 4$$

$$t + 6 = 4$$

$$t = -2$$

Exercise 1E-5

The line has the direction of the normal to the plane. So its parametric equations are

$$x = 1 + t, \quad y = 1 + 2t, \quad z = -1 - t$$

Substituting into the equation of the plane:

$$2(1 + t) - (1 + 2t) + (-1 - t) = 1$$

$$2 + 2t - 1 - 2t - 1 - t = 1$$

$$-t = 1$$

$$t = -1$$

Substituting into the parametric equations gives the intersection point:

$$(0, 1, 0)$$

Exercise 1E-6

Let $P_0 = (x_0, y_0, z_0)$ be a point and let

$$\mathbf{N} = (a, b, c)$$

be a normal vector to the plane.

The distance from the point to the plane is the component of the vector $\overrightarrow{OP_0}$ in the direction of N .

$$\begin{aligned} \text{Distance} &= \frac{|\overrightarrow{OP_0} \cdot \mathbf{N}|}{|\mathbf{N}|} \\ &= \frac{|(x_0, y_0, z_0) \cdot (a, b, c)|}{\sqrt{a^2 + b^2 + c^2}} \end{aligned}$$

$$= \frac{|ax_0 + by_0 + cz_0|}{\sqrt{a^2 + b^2 + c^2}}$$

If the plane equation is

$$ax + by + cz + d = 0$$

then the distance from the point (x_0, y_0, z_0) to the plane is

$$\frac{|ax_0 + by_0 + cz_0 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

$$B + C = \begin{pmatrix} 1 & -1 \\ -1 & 7 \\ 0 & 3 \end{pmatrix}$$

$$B - C = \begin{pmatrix} 1 & -3 \\ 5 & -1 \\ -2 & 1 \end{pmatrix}$$

$$2B - 3C = \begin{pmatrix} 2 & -5 \\ 10 & 18 \\ -8 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} -3 & 1 \\ -3 & -7 \end{pmatrix}$$

$$AC = \begin{pmatrix} 6 & 3 \\ 4 & 6 \end{pmatrix}$$

IF-3

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} a^2 + bc & ab + bd \\ ca + dc & cb + d^2 \end{pmatrix}$$

$$a^2 = -bc, \quad d^2 = -bc$$

$$b(a + d) = 0, \quad c(a + d) = 0$$

Case 1: $a + d \neq 0$

$$b = 0, \quad c = 0$$

$$a = 0, \quad d = 0$$

Case 2: $a + d = 0$

$$d = -a, \quad bc = -a^2$$

Answer:

$$\begin{pmatrix} a & b \\ c & -a \end{pmatrix} \quad \text{where } bc = -a^2$$

IF-4

$$AB \neq BA$$

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix} \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 & 1 \\ 2 & 1 \end{pmatrix}$$

$$BA = \begin{pmatrix} 0 & 1 \\ 1 & 2 \end{pmatrix}$$

$$AB \neq BA$$

$$AB - BA = \begin{pmatrix} 1 & 0 \\ 1 & -1 \end{pmatrix}$$

It is not commutative.

IF-5

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}^3 = \begin{pmatrix} 1 & 2 \\ 2 & 3 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^3 = \begin{pmatrix} 1 & 3 \\ 0 & 1 \end{pmatrix}$$

Guess

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$$

Proof by induction

$$\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}^{n+1} = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & n+1 \\ 0 & 1 \end{pmatrix}$$

IF-6

$$\begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} = \begin{pmatrix} AA & 0 \\ 0 & BB \end{pmatrix}$$

IF-7

$$AB - BA = I_2$$

$$A = \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

$$AB = \begin{pmatrix} 1 & a+b \\ 0 & 1 \end{pmatrix}$$

$$BA = \begin{pmatrix} 1 & a+b \\ 0 & 1 \end{pmatrix}$$

$$AB - BA = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} \neq I_2$$

IF-8

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} a \\ d \\ g \end{pmatrix}$$

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} b \\ e \\ h \end{pmatrix}$$

TG-1

$$A = \begin{pmatrix} 3 & 1 & -1 \\ -1 & 2 & 0 \\ 1 & -1 & 1 \end{pmatrix} \quad b = \begin{pmatrix} 8 \\ 3 \\ 0 \end{pmatrix}$$